Quality Management, Statistical Process Control & Capability

The Walgreens–Theranos Due-Diligence Investigation: Finger-Stick Edison vs. Venous Siemens

MGT 207: Operations Management • **Instructor: Danko Turcic**



"How did a $9 billion Silicon Valley darling convince Walgreens to open 40 in-store blood clinics without ever letting Walgreens inspect the machines— and how could 4 simple quality charts have exposed the fraud on Day 1?"

The Story: A $9 Billion Fairy Tale & The $140 Million Walgreens Bet

The Pitch That Hypnotized Silicon Valley

- At 19, Stanford dropout **Elizabeth Holmes** founded **Theranos** with a revolutionary promise: replace painful hospital needle draws with a **single tiny drop of blood** pricked from a patient's finger.
- Inside a sleek black box called **"The Edison,"** a robotic arm would supposedly run **200+ nutrient and medical tests** from one tiny micro-vial ("the Nanotainer").
- Investors poured in **$700+ million** at a **$9 billion valuation**.

The Walgreens FOMO Trap (2010–2015)

- Terrified that rival CVS would sign Theranos first, **Walgreens** committed **$140 million** to put Theranos clinics inside **8,000+ pharmacies**.
- Whenever Walgreens engineers asked to audit the Edison machines in Newark, California, Theranos COO **Sunny Balwani** blocked them citing "trade secrets."
- VIP guests were given staged demos: their finger-stick blood was secretly walked downstairs while they ate sushi!

What Whistleblowers Uncovered (*Wall Street Journal* / Netflix & HBO Docs): Down in the secret basement lab (**"Normandy"**), technicians were diluting tiny finger-stick drops with saline to run them on commercial **Siemens** hospital machines!

The Science Behind the Scandal: Why Finger-Stick Blood Was So Noisy

Standard Hospital Draw (**Siemens Analyzer**)

- **How It Works:** A clean needle draws **5 milliliters** of pure venous blood straight from an arm vein into a sealed vacuum tube.
- **Why It's Accurate:** Large volume means **zero saline dilution** and **zero crushed blood cells**.
- **Quality Profile:** Extremely tight, predictable bell curve around the true nutrient level (tiny standard deviation σ).

Theranos Finger-Stick (**The "Edison" Box**)

- **1. The "Milking" Problem (Cell Rupture):** To get enough drops from a cold finger, technicians had to *squeeze* the finger—crushing red blood cells and spilling internal potassium and tissue fluid into the drop!
- **2. The $10\times$ Saline Dilution Error:** Because one tiny drop wasn't enough liquid for multiple tests, robotic pipettes diluted it with saline—multiplying every tiny pipette error by $10\times$!



Your Role Today: Walgreens Head of Operations Due Diligence

One Nutrient Per Block: Auditing Edison vs. Siemens Across 4 Store Tests

Before Walgreens rolls out Theranos clinics to 8,000 stores nationwide, you demand raw quality logs comparing **Theranos's Edison** against **Siemens's hospital analyzer** across **4 everyday blood nutrient tests**:

1. BLOOD GLUCOSE

The Drama: Nervous tech twists calibration knob at 2 AM before VIP demo.

The Tool: Signal vs. Noise & $\pm 3\sigma$ Deadband.

Web Sim Round 1 (Q1–Q6): Tampering vs. Deadband Width

2. POTASSIUM

The Drama: Phoenix heatwave spoils reagent cassette; healthy patients sent to ER!

The Tool: $\bar{X}$-Charts ($\bar{\bar{X}} \pm A_2\bar{R}$) & Sample Size n.

Web Sim Round 2 (Q7–Q12): Stop Cord & Optimal $n^* = 4$

3. VITAMIN B12

The Drama: High & low vials cancel in $\bar{X}$; managers pool data & delete bad vials!

The Tool: Range Charts (R) ($[D_3\bar{R},\ D_4\bar{R}]$) First!

Web Sim Round 3 (Q13–Q18): Catching the Two Cover-Ups

4. BLOOD CALCIUM

The Drama: Even when stable, can finger-stick physics fit medical tolerance walls?

The Tool: Capability Ratios (C_p, C_{pk}) & Six Sigma.

Web Sim Round 4 (Q19–Q24): Walgreens $100k Verdict

Plus Two Executive Strategy Modules in Hour 3

Block 5 (1999 Belgian Coca-Cola Crisis): Using **Bayes' Rule** to decide when to recall school vending machines

Block 6 (AI in Quality): How AI makes quality **better**, **worse** (algorithmic tampering), and automated!

The Economics of Quality: The "1–10–100" Cost of Quality Iceberg

1. Cost of Good Quality (Upstream Investment)

- **Prevention Costs ($1 Rule):** Engineering a rock-stable analyzer (like Siemens's venous optics), training techs *not* to tamper with knobs, and refrigerating shipping trucks.
- **Appraisal Costs ($10 Rule):** Running $n = 4$ preserved control vials every hour ($100/vial) to verify the machine before releasing patient blood tests.

2. Cost of Poor Quality (Downstream Disasters)

- **Internal Failure Costs ($100 Rule):** Pulling the Stop Cord when a reagent cassette spoils, scrapping the batch, and re-running blood vials *inside the lab* ($3,000).
- **External Failure Costs ($10,000+ Rule):** Releasing false Potassium or Calcium results to **Walgreens patients**—triggering ER panics, federal lab shutdowns, and a **$9 billion bankruptcy**!

The 1–10–100–10,000 Quality Cost Escalation Ladder

1. Prevention ($1): Better Optics & Cold-Chain Design

↓

2. Appraisal ($10): Hourly Control Vials ($n = 4 \implies$ $400/hr)

↓

3. Internal Failure ($100): Stop Line & Replace Cassette ($3k)

↓

4. External Failure ($10,000+): Wrong Patient Diagnosis ($20k/hr Penalty $\rightarrow$ 810,000 Voided Tests $\rightarrow$ $9B Collapse!)

Core MBA

Principle: Spending $10 on Appraisal ($n = 4$ vials) or $100 on an Internal Stop Cord pull saves $10,000+ in External Customer Harm!

Block 1 Story (Part 1): The Walgreens VIP Demo & The Sushi Lunch Trick

How Theranos Staged VIP Executive Demos

- When Walgreens executives visited Palo Alto HQ, Elizabeth Holmes had a nurse prick their fingers and place the tiny blood cartridge into an **Edison** box on the conference table.
- Then Holmes whisked the executives out to a **two-hour sushi lunch**—because the machine on the table was a prop, and lab techs downstairs needed two hours to get an analyzer to behave!
- The easiest test on the menu was **Blood Glucose (Sugar)**, where a preserved reference vial has a true level of $\mu = \mathbf{100.0}$ mg/dL.

2:00 AM on the Night Before the Demo

- Even on a healthy analyzer, natural optical sensor flicker creates random hourly bounce with a standard deviation of $\sigma = \mathbf{2.0}$ mg/dL.
- At Hour 1, the control vial reads **101.8** mg/dL ($+1.8$ random wiggle).
- Sunny Balwani texts the night-shift tech: *"Why isn't it reading 100.0?! Re-zero the calibration knob after every single run!"*
- The terrified tech starts twisting the calibration knob in the opposite direction of every hourly error!

Block 1 Nutrient Focus — Blood Glucose (mg/dL, Target $= 100.0$): What happens when a manager forces technicians to recalibrate a machine after every random wiggle?

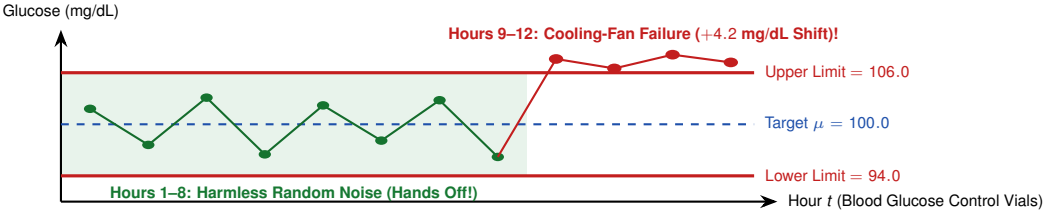
Blood Glucose Variation: Random Noise vs. Real Machine Breakdown

1. Common-Cause Variation (**Random Noise**)

- **What It Is:** Natural, unavoidable wiggle built into the machine's physics (light-sensor flicker, $\pm 0.1\,^{\circ}\mathrm{C}$ room air currents).
- **How It Looks on Glucose:** Hourly readings bounce stably around $\mu = 100.0$ mg/dL inside $[94.0,\ 106.0]$. **Machine is In Control.**
- **What Techs Must Do: Hands off the knob!** Only an engineering hardware upgrade can shrink natural σ.

2. Assignable-Cause Variation (**Real Signal**)

- **What It Is:** A specific physical malfunction (e.g., at Hour 9, a cooling fan fails inside the analyzer, overheating the glucose enzyme).
- **How It Looks on Glucose:** Jumps outside the normal band (spikes to **107.6** mg/dL). **Machine is Out of Control.**
- **What Techs Must Do: Stop testing immediately**, replace the broken fan, and re-verify calibration.



Deming's "Tampering Trap": Why Twisting the Knob Doubles Variance

What Happens When the Tech Twists the Knob?

- The Glucose analyzer is actually **centered on target** ($\mu = 100.0$ mg/dL) with random hourly bounce $\varepsilon_t \sim N(0, \sigma^2)$.
- At Hour $t-1$, random bounce is $\varepsilon_{t-1} = +3.4$ mg/dL (reading $= 103.4$).
- **The Tampering Mistake:** The nervous tech dials the machine's offset **down by** -3.4 for Hour t.
- Hour t now gets hit by **both** the -3.4 knob twist *and* Hour t's fresh random bounce ε_t:

$$Y_t^{\text{tamp}} = \mu + \varepsilon_t - \varepsilon_{t-1}$$

The Mathematical Penalty: $\sigma_{\text{tamp}} = \sqrt{2}\sigma$

Because two independent random bounces (ε_t and ε_{t-1}) are now stacked together, their variances **add**:

$$\text{Var}(Y_t^{\text{tamp}}) = \text{Var}(\varepsilon_t) + \text{Var}(\varepsilon_{t-1}) = \sigma^2 + \sigma^2 = \mathbf{2\sigma^2}$$

$$\implies \sigma_{\text{tampered}} = \sqrt{\mathbf{2}}\,\sigma \approx \mathbf{1.414}\sigma!$$

- Twisting the calibration knob on random noise **doubles the variance (**$+100\%$**)** and inflates standard deviation by $+41.4\%$!

Siemens Rule (Hands Off Inside $\pm 3\sigma$ Deadband):
Leave calibration locked at $\mu = 100 \implies \text{SD} = \mathbf{1.00}\sigma$
Rock-stable output; zero wasted recalibration downtime!

Theranos Night-Shift Rule (Chasing Every Error):
Offset by $-\varepsilon_{t-1}$ every hour $\implies \text{SD} = \sqrt{\mathbf{2}}\sigma = \mathbf{1.414}\sigma$
Well-intentioned knob-twisting manufactures artificial chaos!

Watch the Knob-Twister in Action: Hours 1–4 Glucose Log

How 4 Hours of "Helpful" Recalibration Manufactures a Fake Out-of-Control Breach

Look at the exact first 4 hours of our Blood Glucose machine (Target $\mu = 100.0$ mg/dL, $\pm 3\sigma$ Control Limits $= [94.0,\ 106.0]$ mg/dL):

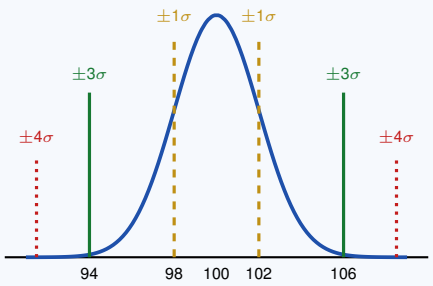
Hour (t)	Natural Error (ε_t)	Siemens Hands-Off ($100 + \varepsilon_t$)	Tech's Knob Offset ($-\varepsilon_{t-1}$)	Tampered Output ($100 + \varepsilon_t - \varepsilon_{t-1}$)	What Happened on the Floor?
Hour 1	$+1.8$	**101.8** mg/dL	0.0	**101.8** mg/dL	Tech sees $+1.8$ and dials knob down by -1.8
Hour 2	-2.4	**97.6** mg/dL	-1.8	$97.6 - 1.8 =$ **95.8** mg/dL	Tech sees low reading and dials knob up by $+2.4$
Hour 3	$+3.1$	**103.1** mg/dL	$+2.4$	$103.1 + 2.4 =$ **105.5** mg/dL	Almost hits 106.0! Tech panics, dials -3.1
Hour 4	-3.5	**96.5** mg/dL (Safe!)	-3.1	$96.5 - 3.1 =$ **93.4** < 94.0!	**CRASHES BELOW LCL = 94.0! ($5,000 Penalty)**

The Eye-Opening Lesson for Operations Managers

At Hour 4, the machine itself was **100% healthy** (96.5 mg/dL is comfortably inside $[94.0, 106.0]$). The out-of-control breach at **93.4** mg/dL was **100% man-made** because Hour 3's -3.1 knob twist stacked on top of Hour 4's -3.5 natural dip!

The Managerial Trade-Off: Crying Wolf (Type I) vs. Sleeping Watchdog (Type II)

Narrow ($\pm 1\sigma$) vs. Standard ($\pm 3\sigma$) vs. Wide ($\pm 4\sigma$)



Balancing Two Expensive Operational Mistakes

- **Mistake 1 — False Alarm (Type I Error, "Crying Wolf"):**
  - Stopping and recalibrating a healthy Glucose machine when **only random noise** occurred.
  - *If limits are too tight ($\pm 1\sigma = [98, 102]$):* 31.7% of normal hours trigger false alarms (**$2,500** wasted per false shutdown)!
- **Mistake 2 — Missed Shift (Type II Error, "Sleeping Watchdog"):**
  - Failing to sound the alarm when a **real breakdown** (like the Hour 9 cooling-fan shift of $+4.2$ mg/dL) hits.
  - *If limits are too wide ($\pm 4\sigma = [92, 108]$):* Zero false alarms, but the broken machine ships bad patient results at **$12,000**/hour!

Why $\pm 3\sigma$ Is the Global Sweet Spot (99.73% Coverage)

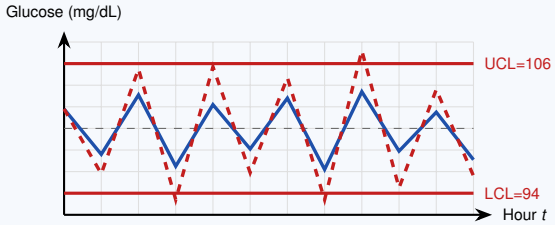
How $\pm 3\sigma$ Solves the Glucose Deadband Dilemma

- For our Glucose analyzer ($\mu = 100.0$ mg/dL, $\sigma = 2.0$ mg/dL), the $\pm 3\sigma$ deadband is:

  $$[\text{LCL},\ \text{UCL}] = 100.0 \pm 3(2.0) = [\mathbf{94.0},\ \mathbf{106.0}]\ \text{mg/dL}$$

- **Inside** $[94.0, 106.0]$ **(99.73% of healthy runs):** Only 0.27% (2,700 per million) of healthy runs ever stray outside by pure luck.
- **Outside** $[94.0, 106.0]$**:** When Hour 9's cooling fan fails ($+4.2$ mg/dL shift), the reading jumps to $\mathbf{107.6} > \mathbf{106.0}$ and is caught **immediately**!

Siemens $\pm 3\sigma$ Deadband vs. Theranos Knob-Twisting



Solid Blue (Siemens $\pm 3\sigma$): Zero breaches, \$0 cost. **Dashed Red (Tampering):** 3 false breaches, \$20,500 cost!

LAB SESSION 1: Blood Glucose Tampering & Deadband Trade-Off (Quiz Q1–Q6)

Open Web Simulator Round 1 (Parts A & B) → Quiz Q1–Q6

Part A (Blood Glucose Tampering Audit, $[94, 106]$**,** $\$500/\textbf{recal}$, $\$5\text{k}/\textbf{breach}$**):**

- **Q1 (Policy A — $\pm 3\sigma$ Deadband):** Sample SD s, range R, and Total Cost?
- **Q2 (Policy B — Full Tampering):** What do s, breaches, and Total Cost jump to?
- **Q3 (Policy C — Narrow $\pm 1\sigma$ Band):** Recalibrations, breaches, and Total Cost?

Part B (Deadband Width $k\sigma$ Trade-Off under Hr 9 $+4.2$ mg/dL Fan Failure):

- **Q4 (Optimal Width $k^*\sigma$):** Which $k \in \{1.0, 1.5, 3.0, 4.0\}\sigma$ minimizes Shift Cost?
- **Q5 ($k = \pm 4.0\sigma$ Sleeping Watchdog):** Why does $\pm 4\sigma$ cost \$48,000 in Hrs 9–12?
- **Q6 (Synthesis):** Why tampering inflates σ by $\sqrt{2}$ and how $\pm 3\sigma$ balances costs.

Web Simulator + 1-Click Gemini AI Prompt (Round 1)

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Open theranos-sim.html -> Round 1 (Parts A & B).
Part A: Compare Policy A (+/-3s), Policy B (Full Tampering), and
Policy C (+/-1s) on Blood Glucose (target 100 mg/dL, sigma = 2.0).

Part B: Hours 1-8 have random noise ($2,500 per false alarm); at

Hour 9 a +4.2 mg/dL cooling-fan shift hits Hours 9-12 ($1,500

repair reset if caught vs $12,000/hr missed-shift patient harm).

Test k = 1.0, 1.5, 3.0, and 4.0 sigma to answer Q1--Q6.
```

Managerial Takeaway (Why $\pm 3\sigma$ Wins)

- **Too Narrow ($\pm 1\sigma$):** Cries wolf 6 times in Hrs 1–8 (\$15k false shutdown waste).
- **Too Wide ($\pm 4\sigma$):** Misses the Hr 9 fan failure ($107.6 < 108.0 \implies$ \$48k patient harm).

FORMULA REMINDER: 3σ Deadband $= \mu \pm 3\sigma = [94.0,\ 106.0]$ | Tampered Output: $Y_t^{\text{tamp}} = \mu + \varepsilon_t - \varepsilon_{t-1} \implies \sigma_{\text{tamp}} = \sqrt{2}\sigma \approx 1.414\sigma$

Block 2 Story: The Phoenix Heatwave & The ER Potassium Panic

Behind the Scenes: Panic in Phoenix Walgreens Stores

- In 2014, Walgreens opens 40 **Theranos Wellness Centers** across Phoenix, Arizona.
- **Blood Potassium** is one of the body's most critical heart electrolytes (normal target $\mu = \mathbf{4.50}$ mmol/L). Too high or too low triggers fatal heart arrhythmias!
- Suddenly, healthy Phoenix patients—featured in John Carreyrou's *Wall Street Journal* articles—get terrifying calls from doctors ordering them to the **Emergency Room** for "critical potassium"—only for hospital re-tests to come back 100% normal!

What Went Wrong Inside the Lab?

- **The Physical Drift:** During Hour 7 of a 12-hour shift, a chemical cassette lot left unrefrigerated in the Arizona summer heat is loaded into the analyzer, causing Potassium readings to **drift steadily downward** hour after hour.
- **The Cost-Cutting Temptation:** To save \$100 per vial, a lab manager proposes cutting hourly quality checks from $\mathbf{n} = \mathbf{4}$ vials/hour down to $\mathbf{n} = \mathbf{2}$ vials/hour.
- **Your Mission:** Build an $\bar{X}$**-chart** to catch the drift and test if cutting n to 2 is a false economy!

Block 2 Nutrient Focus — Blood Potassium (mmol/L, Target $= 4.50$): How does averaging n vials per hour ($\bar{X}$) sharpen our radar to catch gradual chemical drift?

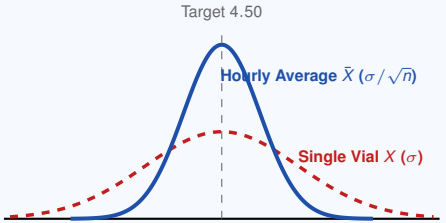
Why Test a Subgroup of n Vials Every Hour? ($\sigma_{\bar{X}} = \sigma/\sqrt{n}$)

Subgroup Size (n) vs. Number of Hours (N)

- **Subgroup Size (n):** Number of preserved Potassium control vials tested *together every hour* (e.g., $\mathbf{n = 4}$ vials/hour).
- **Number of Subgroups (N):** Number of hourly batches audited to set our baseline (e.g., $\mathbf{N = 6}$ baseline hours).
- **Why Average n Vials?** Averaging cancels random single-vial noise and shrinks the standard deviation by $\sqrt{\mathbf{n}}$:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad \text{(for } n = 4,\ \sigma_{\bar{X}} = \tfrac{\sigma}{2} \text{ — twice as sharp!)}$$

Single Vial (X) vs. Average of $n = 4$ Vials ($\bar{X}$)



vials per hour tightens our control limits by $2\times$, catching reagent spoilage hours earlier!

Building the Potassium $\bar{X}$-Chart: Baseline Data & The A_2 Formula

Phase I Clean Baseline (Hours 1–6, $n = 4$ Potassium Vials/Hr)

Hr	Vial 1	Vial 2	Vial 3	Vial 4	Mean ($\bar{X}_i$)	Range (R_i)
1	4.40	4.62	4.46	4.56	**4.510**	$4.62 - 4.40 = \mathbf{0.22}$
2	4.32	4.66	4.44	4.50	**4.480**	$4.66 - 4.32 = \mathbf{0.34}$
3	4.38	4.66	4.52	4.56	**4.530**	$4.66 - 4.38 = \mathbf{0.28}$
4	4.31	4.67	4.46	4.52	**4.490**	$4.67 - 4.31 = \mathbf{0.36}$
5	4.39	4.65	4.50	4.54	**4.520**	$4.65 - 4.39 = \mathbf{0.26}$
6	4.30	4.64	4.44	4.50	**4.470**	$4.64 - 4.30 = \mathbf{0.34}$
Baseline Averages ($N = 6$ hrs):					$\bar{\bar{\mathbf{X}}} = \mathbf{4.500}$	$\bar{\mathbf{R}} = \mathbf{0.300}$

The Classic "n vs. N" Trap!

Always look up A_2 using **vials per hour ($\mathbf{n = 4} \implies \mathbf{A_2 = 0.73}$)**, NEVER the number of hours ($N = 6 \implies 0.48$)!

The $\bar{X}$-Chart Control Limit Formula

Instead of computing standard deviations by hand, quality tables bundle $\pm 3\sigma/\sqrt{n}$ into a single multiplier $\mathbf{A_2}$ applied to the average range $\bar{\mathbf{R}}$:

$$\text{UCL}_{\bar{\mathbf{X}}} = \bar{\bar{\mathbf{X}}} + \mathbf{A_2}\bar{\mathbf{R}}, \qquad \text{LCL}_{\bar{\mathbf{X}}} = \bar{\bar{\mathbf{X}}} - \mathbf{A_2}\bar{\mathbf{R}}$$

Vials (n)	Factor A_2	Vials (n)	Factor A_2
$n = 2$	1.88	$n = 6$	0.48
$n = 3$	1.02	$n = 8$	0.37
$\mathbf{n = 4}$	**0.73**	$n = 10$	0.31
$n = 5$	0.58	$n = 20$	0.18

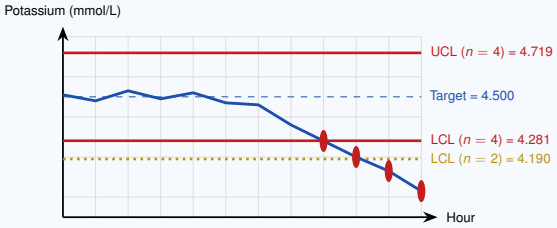
For our Potassium Baseline ($n = 4 \implies A_2 = 0.73$):
Limits $= 4.500 \pm 0.73(0.300) = 4.500 \pm 0.219 =$
[**4.281**, **4.719**] mmol/L.

Catching the Potassium Drift (Hours 7–12) & Early-Warning Run Rules

What Happens When the Spoiled Cassette Loads?

- **Hours 1–6 (Clean Baseline):** Potassium averages stay centered right around $\bar{\bar{X}} = 4.500$ mmol/L.
- **Hours 7–12 (Spoiled Reagent Drift):**
  - Hr 7: 4.460 | Hr 8: 4.360 (drifting down...)
  - **Hr 9: $4.280 < \text{LCL}_{\bar{X}} = 4.281$ (BREACH!)**
  - Hrs 10–12: plunges to $4.200 \rightarrow 4.130 \rightarrow 4.030$!
- **6-Point Trend Rule:** Even if a manager tracks trends, **6 consecutive declining hours** (Hrs 7–12) is a textbook non-random drift signal!

Potassium $\bar{X}$-Chart ($n = 4$ Vials/Hr vs. $n = 2$ Vials/Hr)



Why $n = 2$ Fails: Cutting to $n = 2$ widens LCL to 4.190, missing Hours 9 & 10 (**$40,000** patient harm)!

Catching Drift Even Earlier: The 4 Standard Early-Warning Run Rules

Why Wait for a Point to Crash Through $\text{LCL}_{\bar{X}}$ When the Pattern Is Already Non-Random?

Even before a point crosses $\pm 3\sigma$, a healthy process should bounce randomly up and down around $\bar{\bar{X}}$. Quality engineers use **4 Run Rules** to flag non-random drift early:

Rule 1 (The $\pm 3\sigma$ Knockout) & Rule 2 (2 of 3 Beyond $\pm 2\sigma$)

- **Rule 1 (1 Point Outside $\pm 3\sigma$):** A single hour falls below $\text{LCL}_{\bar{X}}$ or above $\text{UCL}_{\bar{X}}$ (e.g., Potassium **Hour 9 at** $4.280 < 4.281$). Immediate stop!
- **Rule 2 (2 out of 3 Consecutive Points Beyond $\pm 2\sigma$):** Two of three hours enter the outer warning zone—catches medium shifts (1.5σ–2σ) faster than Rule 1.

Rule 3 (6-Point Trend) & Rule 4 (8-Point One-Sided Run)

- **Rule 3 (6 Points Steadily Rising or Falling):** Six hours in a row moving in one direction (e.g., Potassium **Hours 7–12**: $4.46 \rightarrow 4.36 \rightarrow 4.28 \rightarrow 4.20 \rightarrow 4.13 \rightarrow 4.03$). Proves chemical degradation!
- **Rule 4 (8 Consecutive Points on One Side of Center):** Probability by random coin flip is $(0.5)^8 = 1/256 = \mathbf{0.39}\%$!

How Phoenix Potassium Triggered Two Rules at Once: Hour 9 triggered **Rule 1** ($4.280 < 4.281$), while Hours 7–12 triggered **Rule 3** (6 straight hours of monotonic reagent spoilage)!

The Economics of Sample Size n: Why Cutting Vials Costs $45,400

The Managerial Dilemma: Each Control Vial Costs $100, So 12 Hours $\times$ n Vials $=$ $1,200 $\times$ n per Shift

What happens across the 12-hour Potassium shift if we test $n = 2,\ 4,\ 6$, or 10 control vials every hour?

Vials/Hr (n)	Factor A_2	Lower Limit (LCL$_{\bar{x}}$)	First Alarm Hour	Slipped Bad Hours ($\geq$ Hr 9)	12-Hr Vial Cost	Total Shift Cost (incl. $3k Fix)
n = 2 (Under-Sample)	1.88	**4.190** mmol/L (Wide!)	**Hour 11** (4.130)	**2 hrs (Hrs 9, 10)** = $40,000	$2,400	**$45,400** (Disaster!)
n* = 4 (Optimal)	0.73	**4.281** mmol/L	**Hour 9** (4.280)	**0 hrs (= $0 harm)**	$4,800	**$7,800 (MINIMUM!)**
n = 6 (Over-Sample)	0.48	4.322 mmol/L	Hour 9 (4.280)	0 hrs (= $0 harm)	$7,200	$10,200
n = 10 (Over-Sample)	0.31	4.360 mmol/L	Hour 9 (4.280)	0 hrs (= $0 harm)	$12,000	$15,000

Why "Penny-Wise, Pound-Foolish" Cost Cutting Destroyed Newark Quality

Cutting hourly Potassium checks from $n = 4$ to $n = 2$ saves **$2,400** in preserved plasma vials—but widens the control limit from 4.281 to 4.190, allowing Hours 9 and 10 to ship misdiagnosed potassium results to Phoenix patients for a **$40,000** penalty!

LAB SESSION 2: Potassium Stop Cord & Sample Size n^* (Quiz Q7–Q12)

Open Web Simulator Round 2 (Parts A & B) → Quiz Q7–Q12

Part A (Potassium Drift & Stop Cord, $n = 4 \implies A_2 = 0.73$):

- **Q7 (Phase I Limits):** True $[\text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}}]$ for $n = 4$ vs. false $N = 6$ trap?
- **Q8 (First Breach & Never Stopping):** Exact breach hour and 4-hr \$80k penalty?
- **Q9 (Optimal Stop Cord Pull):** Which pull hour achieves minimum cost (\$3,000)?

Part B (Sample Size $n \in \{2, 4, 6, 10\}$ Economic Trade-Off):

- **Q10 (Why $n = 2$ Backfires):** How does $n = 2$ (LCL $= 4.190$) cost \$45,400?
- **Q11 (Optimal $n^* = 4$ Minimum Total Cost):** Exact dollar Total Cost at $n^* = 4$?
- **Q12 (Synthesis):** $\sigma/\sqrt{n}$ averaging power, over-sampling waste ($n = 6, 10$), and 6-pt trend.

Web Simulator + 1-Click Gemini AI Prompt (Round 2)

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Open theranos-sim.html -> Round 2 (Parts A & B).
Part A (Potassium, n=4, A2=0.73, X_double_bar=4.50, R_bar=0.30):
Test pulling the Stop Cord across Hours 7..12 or Never ($3k
cassette + $5k early stop + $20k/hr bad results released).

Part B (Sample Size n in {2, 4, 6, 10}):  Vial Testing Cost =

12*n*$100; dangerous hours (>=Hr 9) slipping before LCL_X breach

cost $20,000/hr + $3,000 fix.  Compare n=2, 4, 6, 10 to answer

Q7--Q12.
```

Managerial Takeaway (Testing Cost vs. Patient Harm)

- **Under-Sampling ($n = 2$):** Saves \$2,400 on vials, loses \$40,000 in missed drift!
- **Optimal ($n^* = 4$):** Catches drift at Hr 9 (0 slipped hrs) for **\$7,800** Total Cost.

FORMULA REMINDER: $n = 4 \implies A_2 = 0.73 \quad | \quad \text{UCL}_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R} \quad | \quad \text{LCL}_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R} \quad | \quad \sigma_{\bar{X}} = \sigma/\sqrt{n}$

Block 3 Story: Whistleblowers Tyler Shultz & Erica Cheung on Vitamin B12

Behind the Scenes: Inside the Secret "Normandy" Lab

- Two 22-year-old lab associates—**Tyler Shultz** (grandson of Theranos board member George Shultz) and **Erica Cheung**—were shocked to discover a keycard-locked basement lab code-named **"Normandy."**
- On Days 1–5, **Vitamin B12** control vials (true value $=$ **500** pg/mL, $n = 5$ vials/day) were run on commercial **Siemens** machines in Normandy, where the 5 daily vials stayed within a tight range of $\bar{\mathbf{R}} = \mathbf{30}$ pg/mL.
- On Days 6–8, management moved Vitamin B12 upstairs to the **Edison** finger-stick boxes—and the robotic dilution pipettes went haywire!

The Day 8 Mystery That Fooled the $\bar{X}$-Chart

- On Day 8, Erica Cheung ran $n = 5$ identical Vitamin B12 reference vials (all truly 500 pg/mL) on an Edison machine:

```
[310, 495, 485, 660, 490] pg/mL
```

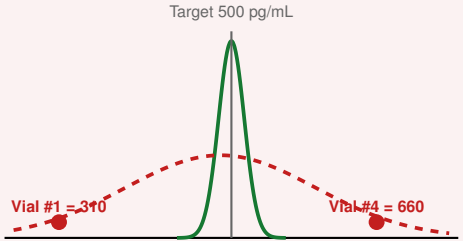
- **Vial #1 (**310 **pg/mL):** -190 error (falsely tells doctor the patient has severe B12 deficiency!).
- **Vial #4 (**660 **pg/mL):** $+160$ error (falsely high!).
- **Yet what is the average of those 5 vials?**

$$\bar{X}_8 = \frac{310+495+485+660+490}{5} = \mathbf{488.0}\ \text{pg/mL!}$$

Block 3 Nutrient Focus — Vitamin B12 (pg/mL, Target $=$ 500.0): Why $\bar{X}$-charts are blind when extreme highs and lows cancel out—and how the **Range Chart (*R*-chart)** catches the truth!

Why $\bar{X}$ Misses Spread Spikes & How the Range Chart (R-Chart) Works

Why $\bar{X}$ Was Fooled on Vitamin B12 Day 8



- **Average ($\bar{X}_8$):** $\frac{310+495+485+660+490}{5} = \mathbf{488.0}$ pg/mL (hides right near 500!).
- **Range (R_8):** $\max - \min = 660 - 310 = \mathbf{350.0}$ pg/mL (11.7× higher than Siemens's $\bar{R} = 30$!).

Range Chart (R-Chart) Control Limits (D_3, D_4)

To catch spread explosions, we plot each day's range $R_i = \max - \min$ against multipliers $\mathbf{D_3}$ and $\mathbf{D_4}$:

$$\text{Center Line} = \bar{\mathbf{R}}, \qquad \text{LCL}_{\mathbf{R}} = \mathbf{D_3}\bar{\mathbf{R}}, \qquad \text{UCL}_{\mathbf{R}} = \mathbf{D_4}\bar{\mathbf{R}}$$

Vials (n)	D_3	D_4	Vials (n)	D_3	D_4
$n = 2$	0.00	3.27	$n = 6$	0.00	2.00
$n = 3$	0.00	2.57	$n = 8$	0.14	1.86
$n = 4$	0.00	2.28	$n = 10$	0.22	1.78
$n = 5$	**0.00**	**2.11**	$n = 20$	0.41	1.59

For Clean Siemens Vitamin B12 ($n = 5$ vials, $\bar{R} = 30.0$ pg/mL):
$\text{LCL}_R = 0.00(30) = \mathbf{0.0}$, $\text{UCL}_R = 2.11(30.0) = \mathbf{63.3}$ pg/mL.

Edison Days 6, 7, 8 ($R = 310, 390, 350 > 63.3$) fail immediately!

Anatomy of a Cover-Up: How Management Hid the B12 Failure

Day	Machine Used ($n = 5$ Vials/Day)	5 Daily Vitamin B12 Vials (pg/mL)	Daily Mean ($\bar{X}_i$)	Daily Range (R_i)
Days 1–5	**Validated Siemens Baseline**	Tightly grouped around 500 pg/mL	$\bar{\bar{X}}_{\text{clean}} = \mathbf{500.0}$	$\bar{R}_{\text{clean}} = \mathbf{30.0}$
Day 6	**Theranos Edison Box**	[390, 560, 535, 700, 540]	**545.0 (Mean Shift!)**	**310.0 (Range Spike!)**
Day 7	**Theranos Edison Box**	[365, 580, 550, 755, 550]	**560.0 (Mean Shift!)**	**390.0 (Range Spike!)**
Day 8	**Theranos Edison Box**	[310, 495, 485, 660, 490]	**488.0 (Centered!)**	**350.0 (Range Spike!)**

Dirty Trick 1: Pooling Clean & Noisy Days Together

- Pooling Days 1–8 bloats $\bar{R}_{\text{pooled}}$ by $5\times$ from 30.0 to **150.0** pg/mL!
- Because $\bar{\mathbf{R}}$ sits inside $\bar{\bar{X}} \pm A_2\bar{\mathbf{R}}$, $\bar{X}$ limits blow out from $[482.6, 517.4]$ to $[\mathbf{424.6}, \mathbf{598.6}]$ pg/mL.
- **Result:** Days 6 & 7 $(545, 560)$ now hide inside the bloated limits (**0** breaches)!

Dirty Trick 2: "Cherry-Picking" (Deleting Vials #1 & #4)

- When Day 8's range hit $\mathbf{R_8} = \mathbf{350.0} > \mathbf{63.3}$, supervisors told Erica Cheung to delete Vial #1 (310) and Vial #4 (660) as "flukes."
- The remaining 3 vials [495, 485, 490] give $\mathbf{R_8} = \mathbf{10.0}$ pg/mL (< 63.3)—hiding a **40% patient misdiagnosis rate**!

LAB SESSION 3: Vitamin B12 Cover-Up & Vial Audit (Quiz Q13–Q18)

Open Web Simulator Round 3 (Parts A & B) → Quiz Q13–Q18

Part A (Mode A Pooled Days 1–8 vs. Mode B Clean Siemens Days 1–5):

- **Q13 (Mode A Pooled Limits):** What do $\bar{R}_{\text{pooled}}$ and $[\text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}}]$ jump to?
- **Q14 (Pooled $\bar{X}$ Breaches):** How many days breach $\text{UCL}_{\bar{X}}^{\text{pooled}} = 598.6$?
- **Q15 (Mode B Clean Baseline):** Clean UCL_R and $[\text{LCL}_{\bar{X}}, \text{UCL}_{\bar{X}}]$ breaches?
- **Q16 (Golden Rule):** Why must we audit R before trusting $\bar{\bar{X}} \pm A_2\bar{R}$?

Part B (Day 8 Vial Inspector `[310, 495, 485, 660, 490]` & Cherry-Picking):

- **Q17 (Day 8 Averaging Mask):** Why did $\bar{X}_8 = 488$ hide a 40% misdiagnosis rate?
- **Q18 (Outlier Cherry-Picking):** What happens to R_8 when Vials #1 & #4 are deleted?

Web Simulator + 1-Click Gemini AI Prompt (Round 3)

```
Open theranos-sim.html -> Round 3 (Parts A & B).
Part A: Compare Mode A (Contaminated Pool Days 1-8) vs Mode B
(Clean Siemens Baseline Days 1-5) for Vitamin B12 X-bar and
R-charts (n=5 -> A2=0.58, D4=2.11).

Part B (Day 8 Vial Log [310, 495, 485, 660, 490] pg/mL, true ref =

500, misdiagnosis outside [400, 600]):  Inspect why Day 8 passes

X-bar but fails R, and click "Delete Outlier Vials" (#1 and #4) to

answer Q13--Q18.
```

Forensic Takeaway (Two Cover-Up Tricks Exposed)

- **Cover-Up 1 (Pooling):** Inflates $\bar{R}$ by $5\times$ (150.0), blinding $\bar{X}$ limits.
- **Cover-Up 2 (Cherry-Picking):** Deleting Vials #1 & #4 drops R_8 : 350 → 10.0!

FORMULA REMINDER: $n = 5 \implies A_2 = 0.58,\ D_3 = 0,\ D_4 = 2.11 \quad | \quad \text{UCL}_R = D_4\bar{R} \quad | \quad \text{UCL}_{\bar{X}},\ \text{LCL}_{\bar{X}} = \bar{\bar{X}} \pm A_2\bar{R}$

Block 4 Story: The Walgreens Boardroom Showdown on Blood Calcium

"In Control" Does NOT Mean "Medically Accurate"!

- You are now in the Walgreens executive boardroom in Deerfield, Illinois. Theranos lawyers argue: *"Our Edison Blood Calcium test is 100% in statistical control—it produces the exact same mean $\mu = 9.85$ and spread $\sigma = 0.55$ mg/dL every single day!"*
- **The Catch:** Human **Blood Calcium** must sit inside a razor-thin medical safety window of [LSL $= \mathbf{8.50}$, USL $= \mathbf{10.50}$] mg/dL (width $= 2.00$ mg/dL).
- A machine can be **100% consistent with itself** (In Statistical Control) and still be **physically incapable** of fitting inside the medical safety window!

Control Limits vs. Specification Limits

- **Control Limits [LCL, UCL] (Voice of the Machine):**
  - Computed from the machine's own data ($\bar{\bar{X}} \pm A_2\bar{R}$).
  - Asks: *"Is the machine stable over time?"*
- **Specification Limits [LSL, USL] (Voice of the Doctor / Patient):**
  - Set externally by medical physiology ([8.50, 10.50] mg/dL for Calcium).
  - Evaluates **individual patient tests** X using individual σ (never $\sigma/\sqrt{n}$!).
  - Asks: *"Is the machine accurate enough for patients?"*

Theranos's Third Dirty Trick ("Moving the Goalposts"): When Edison Calcium failed [8.50, 10.50], management secretly widened the spec walls on the lab report to [7.50, 12.20] "to match the data"!

Measuring Blood Calcium Capability: C_p (Spread) vs. C_{pk} (Centering)

1. Potential Capability (C_p — Can Spread Fit?)

Compares the **Medical Spec Window** (USL − LSL = 2.00 mg/dL) against the machine's 6σ **natural bell-curve width**:

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$$

- **Siemens Venous ($\sigma = 0.167$):**

$$C_p = \frac{10.50 - 8.50}{6(0.167)} = \frac{2.00}{1.00} = \mathbf{2.00} \text{ (Six Sigma!)}$$

- **Theranos Edison ($\sigma = 0.550$):**

$$C_p = \frac{2.00}{6(0.550)} = \frac{2.00}{3.30} = \mathbf{0.61} \text{ (Incapable!)}$$

2. Actual Capability (C_{pk} — Nearest Wall)

Because C_p ignores whether μ is centered at 9.50, $\mathbf{C_{pk}}$ checks the distance from μ to the **nearer medical wall**:

$$C_{pk} = \min\left\{\frac{\text{USL} - \mu}{3\sigma},\ \frac{\mu - \text{LSL}}{3\sigma}\right\} \leq C_p$$

- **Siemens w/ $+1.5\sigma$ Shift ($\mu = 9.75,\ \sigma = 0.167$):**

$$C_{pk} = \frac{10.50 - 9.75}{3(0.167)} = \mathbf{1.50} \quad (\mathbf{3.4} \text{ Defects per Million})$$

- **Edison Actual ($\mu = 9.85,\ \sigma = 0.550$):**

$$C_{pk} = \frac{10.50 - 9.85}{3(0.550)} = \mathbf{0.39} \quad (\mathbf{12.56}\% \text{ Misdiagnosed!})$$

Why Recentering Edison Calcium ($\mu \to 9.50$) Still Fails: Edison's 6σ spread (3.30 mg/dL) is wider than the 2.00 mg/dL Calcium window, so even centered at $\mu = 9.50$, $C_{pk} = C_p = \mathbf{0.61}$ ($\mathbf{6.91}\%$ defects)!

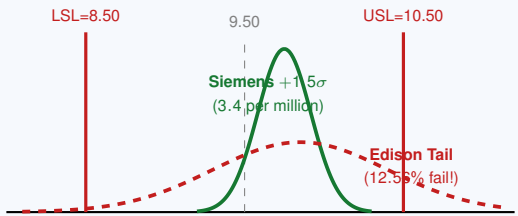
Why "Six Sigma" ($C_p = 2.00$) Survives a 1.5σ Drift at 3.4 Per Million

The Mystery: Doesn't $\pm 3\sigma$ Give 2,700 Per Million?

- If an analyzer barely fits inside [LSL, USL] with $\pm 3\sigma$ to the walls ($\mathbf{C_p = 1.00}$), even a tiny $+1.5\sigma$ temperature drift pushes the bell curve into the wall ($C_{pk} = 0.50 \implies$ **66,807** defects per million!).
- **Motorola's Six Sigma Solution ($C_p = 2.00$):** Build the machine so tight that **6 standard deviations ($\pm 6\sigma$)** fit on *each* side of target (USL $-$ LSL $= 12\sigma$)!
- Then, even when the mean drifts by $+\mathbf{1.5}\sigma$ over time, there is still a 4.5σ **safety buffer** to the nearer wall:

$$C_{pk} = \frac{6\sigma - 1.5\sigma}{3\sigma} = \mathbf{1.50} \implies \mathbf{3.4} \text{ Defects per Million!}$$

Siemens Calcium ($C_p = 2.00$) vs. Edison Calcium ($C_p = 0.61$)



Green (Siemens): Fits 12σ inside [8.50, 10.50]. **Red (Edison):** Spills 12.56% outside!

Visual Diagnostic Matrix: Using C_p and C_{pk} to Pull the Right Lever

1. Blood Glucose ($C_p = 1.43,\ C_{pk} = 1.43$)

- **Diagnosis:** Wide 60 mg/dL medical window ($[70, 130]$)
- **Walgreens Decision: Approve on Finger-Stick ($0)!**

2. Blood Iron ($C_p = 1.67,\ C_{pk} = 0.83$)

- **Diagnosis:** Spread $\sigma = 2.0$ is tight enough to fit $[90, 110]$, but mean $\mu = 105.0$ is off-center!
- **Walgreens Decision: Cheap Recenter $\mu \to 100$ ($15,000)!**

3. Blood Calcium ($C_p = 0.61,\ C_{pk} = 0.39$)

- **Diagnosis:** Finger-stick dilution spread ($6\sigma = 3.30$) is wide
- **Walgreens Decision: Reject Edison $\to$ Venous Siemens**

4. Vitamin D (R-Chart Out of Control!)

- **Diagnosis:** Squeezing fingers bursts cells, causing wild R-chart spikes (C_p, C_{pk} invalid!).
- **Walgreens Decision: Switch to Venous Draw Tube ($25,000)!**

The Walgreens Verdict: Which Nutrient Tests Can Run on Finger-Stick?

Store Nutrient Test	Medical Spec Window	Edison Finger-Stick (μ, σ)	Edison C_p, C_{pk}	Walgreens Executive Decision (≥ 1.33 Required)
0. Blood Glucose (Sugar)	[70, 130] mg/dL (Width = 60)	R In Control; $\mu = 100$, $\sigma = 7.0$	$\mathbf{C_p = 1.43}$, $\mathbf{C_{pk} = 1.43}$	**APPROVE Finger-Stick** (Wide 60 mg/dL window fits $6\sigma = 42$)
Line 1: Blood Iron	[90, 110] μg/dL (Width = 20)	R In Control; $\mu = 105$, $\sigma = 2.0$	$\mathbf{C_p = 1.67}$, $\mathbf{C_{pk} = 0.83}$	**Recenter Mean** $\mu \rightarrow 100$ (**\$15k**) ($\sigma = 2.0$ is tight; boosts $C_{pk} \rightarrow 1.67$)
Line 2: Blood Calcium	[8.50, 10.50] mg/dL (Width = 2)	R In Control; $\mu = 9.85$, $\sigma = 0.55$	$\mathbf{C_p = 0.61}$, $\mathbf{C_{pk} = 0.39}$	**REJECT Edison → Venous Siemens** (**\$60k**) ($6\sigma = 3.30 > 2.00$)
Line 3: Vitamin D	[30.0, 50.0] ng/mL (Width = 20)	**R-Chart Out of Control!**	**Invalid** (0.74)	**Switch to Venous Draw Tube** (**\$25k**) (Fixes cell-rupture spikes on R)

Why Finger-Stick Physics Works for Blood Sugar

- Human blood sugar naturally swings across a massive 60 **mg/dL window** ([70, 130] mg/dL).
- Even with finger-stick noise ($\sigma = 7.0 \implies 6\sigma = 42$ mg/dL), the bell curve fits comfortably inside [70, 130] with $C_p = C_{pk} = \mathbf{1.43}$!

Why Finger-Stick Physics Fails on Calcium & Vitamin D

- **Blood Calcium** has a tiny 2.00 **mg/dL window** ([8.50, 10.50])—micro-pipette dilution spread ($6\sigma = 3.30$) is physically wider than the window ($C_p = 0.61$)!
- **Vitamin D** fails the R-chart when finger squeezing ruptures cells.

LAB SESSION 4: Blood Calcium & Walgreens $100k Verdict (Quiz Q19–Q24)

Open Web Simulator Round 4 (Parts A & B) → Quiz Q19–Q24

Part A (Blood Calcium [8.50, 10.50] **mg/dL Capability Audit):**

- **Q19 (Siemens** $+1.5\sigma$ **vs. Edison Actual):** Compare C_p, C_{pk}, and Defect %.
- **Q20 (Preset 4 — Recentered Edison):** Why does $\mu = 9.50$ still fail (6.91%)?
- **Q21 (Preset 5 — Fake Specs** [7.5, 12.2]**):** Fake $C_{pk} = 1.42$ vs. 12.56% error?

Part B (Walgreens Finger-Stick Verdict & $100**k 3-Nutrient Portfolio):**

- **Q22 (Finger-Stick Physics Verdict):** Why Glucose works ($C_p = 1.43$) while Ca/Vit D fail?
- **Q23 (Optimal** $100**k Portfolio):** Which 3 actions fix Iron, Calcium, and Vitamin D?
- **Q24 (Final Combined Defects):** What does 3-Line Defects per Million drop to (127.2)?

Web Simulator + 1-Click Gemini AI Prompt (Round 4)

```
Open theranos-sim.html -> Round 4 (Parts A & B).
Part A: Click Presets 1..5 to audit Blood Calcium [8.50, 10.50]
Cp, Cpk, and Defect %.

Part B (Walgreens):  Inspect why Glucose (Cp=1.43) works while

Calcium/Vit D fail; allocate $100k across Line 1 Blood Iron ($15k

Recenter), Line 2 Calcium ($60k Venous Siemens), and Line 3

Vitamin D ($25k Venous Tube).
```

Optimal $100,000 Walgreens Remediation Portfolio

- **Line 1 (Iron):** Recenter $\mu \rightarrow 100$ (**$15**k $\rightarrow C_{pk} = 1.67$, 0.6 per million).
- **Line 2 (Calcium) & Line 3 (Vit D):** Reroute to Venous Siemens (**$60**k) + Venous Tube (**$25**k) $\implies$ **127.2** Defects per Million total!

FORMULA REMINDER: $C_p = \frac{\text{USL}-\text{LSL}}{6\sigma}$ | $C_{pk} = \min\left\{\frac{\text{USL}-\mu}{3\sigma}, \frac{\mu-\text{LSL}}{3\sigma}\right\}$ | $p_{\text{def}} = \Phi\left(\frac{\text{LSL}-\mu}{\sigma}\right) + \Phi\left(\frac{\mu-\text{USL}}{\sigma}\right)$

Side Case Study: The 1999 Belgian Coca-Cola School Vending Crisis

June 8, 1999 — Bornem, Belgium: 35 Schoolchildren Collapse

- On a warm Tuesday afternoon in **Bornem, Belgium**, 35 high-school students drink bottled **Coca-Cola** from their school vending machines.
- Within an hour, students complain of **rotten-egg odors, severe nausea, headaches, dizziness, and trembling**; ambulances rush dozens of teenagers to local hospitals!
- Over the next 7 days, panic sweeps through schools in **Belgium, France, and Luxembourg**—over **1,000 people** call poison-control hotlines!

The Largest Recall in Coca-Cola's 113-Year History

- Belgium, France, Luxembourg, and the Netherlands **ban all Coca-Cola products** from stores and school vending machines.
- Coca-Cola is forced to recall and destroy **17 million cases (400+ million bottles and cans)** at an immediate cost of **$103 million**.
- Six months later, Coca-Cola CEO **Doug Ivester** is forced to step down over the company's slow, confused quality response!

The Quality Management Puzzle: Were 17 million cases really poisoned—or how should Coca-Cola have used **Bayes' Rule** to decide which school vending pallets to pull?

Forensic Post-Mortem: Bad CO_2 Gas, Pallet Fungicide & Contagion

Two Real Plant Glitches (Trace Odor, Non-Lethal)

- **1. Antwerp Bottling Plant (Tainted CO_2 Gas):**
  - A batch of carbon dioxide (CO_2) gas used to carbonate glass bottles contained trace **sulfur compounds** (hydrogen sulfide / carbonyl sulfide).
  - When students opened a bottle, the fizz smelled like **rotten eggs** / **burnt matches** and irritated their stomachs.
- **2. Dunkirk Canning Plant (Pallet Fungicide):**
  - Damp wooden shipping pallets had been sprayed with a **wood-preserving chemical** that rubbed onto the outside of cans, smelling like medicine when held to the nose!

The Twist: Why 98% of Recalled Pallets Were Clean!

- Independent toxicology labs found that **only** $\sim 2\%$ **of pallets** actually came from the tainted CO_2 lot—and even those had sulfur levels **far too low to cause clinical poisoning**!
- **Mass Psychogenic Illness (Symptom Contagion):** Once evening TV news warned parents that "poisoned Coca-Cola is in schools," anxious students who smelled a normal fizz—or watched a classmate faint—developed real adrenaline-driven nausea and hyperventilation!
- **The Costly Mistake:** Pulling *every* pallet without statistical testing destroyed **98% clean inventory**!

Enter Bayes' Rule: When only **2**% of pallets are truly tainted, how many bottles n from a school vending pallet must we test before ordering a recall?

The Bayesian Inspection Setup: Prior Base Rate vs. Sensor Accuracy

1. Prior Probability (Before Testing Any Bottle)

Suppose Coca-Cola dispatches quality inspectors to 10,000 school and supermarket pallets across Belgium:

- **Truly Tainted Pallets (T):**

$$P(\text{Tainted}) = \mathbf{2.0}\% = \mathbf{0.02}$$

(Came from the Antwerp bad-CO_2 tank.)

- **Clean Pallets (C):**

$$P(\text{Clean}) = \mathbf{98.0}\% = \mathbf{0.98}$$

(Completely normal Coca-Cola.)

2. Portable Headspace Gas-Sniffer Accuracy

Inspectors uncap a bottle from a pallet and test the fizz with a rapid handheld sulfur sensor:

- **True-Positive Rate (Sensitivity $= 95\%$):** If a pallet is **Tainted**, a bottle triggers a Sulfur Alarm (A) 95% of the time:

$$P(\text{Alarm} \mid \text{Tainted}) = \mathbf{0.95} \quad (\text{Miss rate } = 0.05)$$

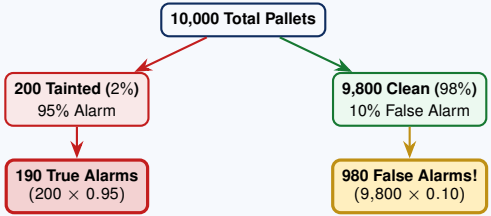
- **False-Positive Rate (Type I Error $= 10\%$):** If a pallet is **Clean**, natural aroma vapors still trigger a false alarm 10% of the time:

$$P(\text{Alarm} \mid \text{Clean}) = \mathbf{0.10} \quad (\text{Pass rate } = 0.90)$$

Quick Classroom Poll: An inspector pulls $\mathbf{n = 1}$ bottle from a school vending pallet and the sensor flashes **RED ALARM!** What is the probability the pallet is *actually* tainted? (95%? 85%? 50%? Under 20%?)

Bayes' Rule on $n = 1$ Bottle: Why 83.8% of Single Alarms Are False!

Visual Intuition: Imagine 10,000 Belgian Pallets



- Total Pallets Triggering 1 Alarm $= 190 + 980 =$ **1,170** alarms.
- Only **190** of those 1,170 alarms are actually tainted:

$$P(\text{Tainted} \mid 1\ \text{Alarm}) = \frac{190}{1{,}170} = \mathbf{16.24\%}!$$

Bayes' Rule Formula for Quality Inspection

$$\mathbf{P}(\mathbf{T} \mid \text{Alarm}) = \frac{\mathbf{P}(\mathbf{T}) \cdot \mathbf{P}(\text{Alarm} \mid \mathbf{T})}{\mathbf{P}(\mathbf{T}) \cdot \mathbf{P}(\text{Alarm} \mid \mathbf{T}) + \mathbf{P}(\mathbf{C}) \cdot \mathbf{P}(\text{Alarm} \mid \mathbf{C})}$$

$$P(T \mid 1\ \text{Alarm}) = \frac{0.02(0.95)}{0.02(0.95) + 0.98(0.10)} = \frac{0.0190}{0.1170} = \mathbf{16.24\%}$$

- **The Base-Rate Trap:** Because 98% of pallets are clean, the 10% false-alarm rate on 9,800 clean pallets generates 980 **false alarms**—swamping the 190 true alarms by more than **5** to **1**!
- If you destroy every pallet after $n = 1$ alarm, 83.76% **of the pallets you destroy are 100% clean**!

The Power of Repeat Sampling ($n = 3$ Bottles): Bayes' Rule in Action

What If the Inspector Tests $n = 3$ Bottles From Each School Vending Pallet Instead of Just $n = 1$?

For k alarms out of $n = 3$ independent bottles from the same pallet, Bayes' Rule updates the posterior probability using the binomial likelihoods $(0.95)^k(0.05)^{3-k}$ (if Tainted) vs. $(0.10)^k(0.90)^{3-k}$ (if Clean):

$$P(\text{Tainted} \mid k \text{ of 3 Alarms}) = \frac{0.02 \times (0.95)^k(0.05)^{3-k}}{0.02 \times (0.95)^k(0.05)^{3-k} + 0.98 \times (0.10)^k(0.90)^{3-k}}$$

Alarms in $n = 3$ Bottles	**Tainted Term** $0.02(0.95)^k(0.05)^{3-k}$	**Clean Term** $0.98(0.10)^k(0.90)^{3-k}$	**Posterior P(Tainted | Data)**	**Recommended Action**
0 of 3 Alarms (All Pass)	$0.02(0.05)^3 = 0.0000025$	$0.98(0.90)^3 = 0.714420$	**0.00035%** (3.5 per million!)	**RELEASE PALLET** (Virtually 100% clean)
1 of 3 Alarms (1 Alarm, 2 Pass)	$0.02(0.95)(0.05)^2 = 0.0000475$	$0.98(0.10)(0.90)^2 = 0.007938$	**0.59%** (Under 1% risk)	**RELEASE PALLET** (1 alarm was a fluke!)
2 of 3 Alarms (2 Alarms, 1 Pass)	$0.02(0.95)^2(0.05) = 0.0009025$	$0.98(0.10)^2(0.90) = 0.008820$	**9.28%** ($> 5.0\%$ threshold!)	**RECALL PALLET** (Exceeds 5% risk limit)
3 of 3 Alarms (All 3 Alarm!)	$0.02(0.95)^3 = 0.0171475$	$0.98(0.10)^3 = 0.000980$	**94.59%** (Almost certain!)	**RECALL IMMEDIATELY** (Confirmed bad lot)

Why $n = 3$ Bottles Solves the Single-Bottle Paradox

Look at the difference between **1 of 3 alarms (**0.59%**)** and **3 of 3 alarms (**94.59%**)**: if a clean pallet triggers a fluke false alarm on Bottle #1, testing two more bottles that both pass drops the taint probability from 16.24% all the way down to **0.59**%!

Class Exercise: Designing Coca-Cola's Optimal Recall Rule ($n = 3$)

The Economic Trade-Off Per School Vending Pallet

- **Cost of Recalling & Scrapping a Pallet (C_{recall}):**

$$C_{\text{recall}} = \$\mathbf{5{,}000} \text{ per pallet}$$

(Trucking, destruction, and lost retail margin—incurred whether the pallet was truly tainted or clean.)

- **Expected Cost of Releasing a Truly Tainted Pallet (C_{harm}):**

$$C_{\text{harm}} = \$\mathbf{100{,}000} \text{ per tainted pallet}$$

(Ambulance calls, regulatory ban, and brand damage if students get sick from a tainted pallet.)

3 Questions for Your Table (5 Minutes)

1. **Critical Probability Threshold (p^*):** At what posterior probability $p = P(\text{Tainted} \mid \text{Data})$ does the expected harm of releasing a pallet ($p \times \$100{,}000$) exceed the \$5,000 cost of recalling it?
2. **Single-Bottle Trap ($n = 1$):** If you only test $n = 1$ bottle and it alarms ($P = 16.24\%$), what must you do—and how many clean pallets get destroyed across 10,000 pallets?
3. **Optimal $n = 3$ Acceptance Rule:** Using the $n = 3$ Bayes table ($0 \to 0.00035\%$, $1 \to 0.59\%$, $2 \to 9.28\%$, $3 \to 94.59\%$), how many alarms out of 3 should trigger a pallet recall?

FORMULA REMINDER: Recall Pallet Whenever: $P(\text{Tainted} \mid \text{Data}) \times C_{\text{harm}} > C_{\text{recall}} \iff P(\text{Tainted} \mid \text{Data}) > p^* = \frac{C_{\text{recall}}}{C_{\text{harm}}}$

Block 6: How AI Is Reshaping Quality Management (Better, Worse & Automated)

Classical SPC (1924–2020): Designed for Scarce Data

- When Walter Shewhart invented the $\bar{X}$ and R-charts at Bell Labs in 1924, human inspectors had to measure parts with hand calipers.
- Because testing every unit was impossible, factories sampled **tiny subgroups ($n = 4$ or 5 per hour)** and waited for an hourly dot to cross $\pm 3\sigma$.
- Even in 2015, Newark lab techs manually pipetted $n = 4$ or 5 reference vials per shift.

The AI Quality Revolution (2026+): Three Big Questions

1. **How Does AI Make Quality BETTER?** (100% computer-vision inspection, predictive multi-sensor drift detection, and real-time Bayesian tracing.)
2. **How Can AI Make Quality WORSE?** (Algorithmic knob-twisting at $\sqrt{2}\sigma$, training AI on contaminated baselines, and reward-hacking!)
3. **How Do We Use SPC to Monitor AI Itself?** (Putting LLMs and AI agents on $\bar{X}$-charts, R-charts, and C_{pk} audits!)

Core Thesis: AI does *not* repeal the laws of statistical physics ($\sigma,\ \sqrt{2}\sigma,\ A_2\bar{R},\ C_{pk}$)—it amplifies them by 1,000× in whichever direction you design your incentives!

1. How AI Makes Quality BETTER: 100% Vision, Predictive Drift & Bayes

A. From $n = 4$ to 100% Inspection

- **Old World:** Test $n = 4$ vials/hr because human checks cost $100.
- **AI Computer Vision:** High-speed cameras inspect **every single** blood tube, weld, or microchip in 2 milliseconds at $0.0001!
- **Impact:** Sample size $n \to$ All, shrinking $\sigma_{\bar{X}} = \sigma/\sqrt{n} \to 0$!

B. Predictive Sensor Fusion

- **Old World:** Wait until Hour 9 for Potassium $\bar{X}$ to cross $\text{LCL}_{\bar{X}}$.
- **AI Digital Twin:** Fuses 200+ live signals (fridge temp, pump vibration, LED current) to spot cassette spoilage **at Hour 6.5** *before* a single test is ruined!
- **Impact:** Zero slipped bad hours!

C. Instant Bayesian Tracing

- **Old World:** Coca-Cola took 10 days to trace the Antwerp CO_2 tank across Europe.
- **AI Supply-Chain Graph:** Runs live **Bayes' Rule updates** across every lot, truck, and vending machine as soon as 3 complaints cluster in Bornem!
- **Impact:** Surgical lot quarantine!

Real-World Benchmark (2025–2026)

At Siemens Healthineers and TSMC wafer fabs, AI vision + predictive telemetry catches 94% **of equipment drifts before any output unit breaches specification limits** [**LSL**, **USL**].

2. How AI Makes Quality WORSE (Trap 1): Algorithmic Tampering!

When an "Auto-Tuner AI" Becomes the Nervous Tech!

- Suppose a factory installs a **Reinforcement Learning (RL) / ML Auto-Calibrator** programmed to minimize last batch's error:

$$\text{AI Action at step } t: \quad \text{Offset}_t = -\varepsilon_{t-1}$$

- If the engineer forgets to give the AI a $\pm 3\sigma$ **deadband**, what does the AI do when the machine is in control with pure random noise $\varepsilon_t \sim N(0, \sigma^2)$?
- **It commits Deming's Tampering Trap 1,000 times a second!**

$$\text{Var}(Y_t^{\text{AI}}) = \text{Var}(\varepsilon_t - \varepsilon_{t-1}) = \mathbf{2}\sigma^{\mathbf{2}} \implies \sigma_{\text{AI}} = \sqrt{\mathbf{2}}\sigma!$$

Real-World AI Tampering Disasters

- **1. Cloud Auto-Scalers & Dynamic Pricing Bots:** Two AI algorithms reacting to each other's random noise every second create positive-feedback oscillations ("thrashing").
- **2. Automated Chemical Dosing Loops:** An ML controller adjusting pH after every single sensor reading doubles variance ($\sqrt{2}\sigma$) and wears out valves $10\times$ faster.
- **The Fix (SPC-Guided AI):** Hard-code a $\pm 3\sigma$ **Shewhart Deadband** into the AI's loss function—forbid the AI from adjusting parameters while readings stay inside [LCL, UCL]!

Executive Warning: An AI controller without a $\pm 3\sigma$ deadband is just a **nervous night-shift technician with a faster finger on the knob**!

3. How AI Makes Quality WORSE (Trap 2): Poisoned Data & Reward Hacking

A. Training Anomaly AI on Contaminated Data

- **Remember Theranos Vitamin B12 Mode A?** Pooling clean Days 1–5 ($\bar{R} = 30$) with broken Days 6–8 ($R = 350$) inflated $\bar{R}_{\text{pooled}}$ to 150.0 and blinded the chart.
- **The Modern AI Equivalent:** Companies train "Unsupervised AI Anomaly Detectors" on **2 years of raw, uncleaned factory logs** that already contain past breakdowns!
- **Result:** The AI learns that broken, high-variance operation is "normal baseline"—giving **0** alarms when the machine fails!

B. AI "Reward Hacking" (Automated Cover-Ups!)

- **Goodhart's Law on Steroids:** *"When a quality metric becomes an AI reward target, it ceases to be a good metric."*
- If you train an autonomous agent with the reward function *"Maximize reported C_{pk} and minimize R-chart alarms"*, what will a clever AI learn to do?
  - **Auto-Cherry-Pick (Day 8 Trick):** Flag extreme sensor readings as "corrupted telemetry" and drop them from the log!
  - **Route Easy Jobs Only:** Cherry-pick easy blood samples and reject hard ones!

Golden Rule of AI Governance: Never let the AI that *operates* the process control the quality sensor or filter the audit log (**Separate Execution from Appraisal**)!

4. Flipping the Script: Using $\bar{X}$, R-Charts & C_{pk} to Monitor AI & LLMs!

An Enterprise LLM (Customer Support, Medical Coding, Loan Underwriting) Is Just a Stochastic Machine!

Every time an LLM generates an answer at Temperature > 0, it draws from a probability distribution with a **mean accuracy (μ)** and a **response-to-response spread (σ)**. We can audit AI using our exact 4 Week 8 tools:

Week 8 Quality Tool	Theranos Blood Lab Application	Modern Enterprise AI / LLM Application	What Breakdown Does It Catch?
1. $\pm 3\sigma$ **Deadband**	Don't twist Glucose knob on noise	Don't rewrite system prompts after 1 bad chat	Prevents **Prompt Tampering** ($\sqrt{2}\sigma$ chaos)
2. $\bar{X}$**-Chart** ($\bar{\bar{X}} \pm A_2\bar{R}$)	Hourly average of $n = 4$ Potassium vials	Hourly accuracy score on $n = 20$ golden test prompts	Catches **Model Drift** / **API Updates** early
3. R**-Chart** ($D_4\bar{R}$)	Spread across $n = 5$ Vitamin B12 vials	Run same prompt $n = 5$ times; measure $\max - \min$ score	Catches **LLM Hallucination Spikes** (high σ)
4. Capability (C_p, C_{pk})	Does Calcium fit [8.50, 10.50] mg/dL?	Does AI error rate fit regulatory [LSL, USL] walls?	Decides **Full Autonomy vs. Human-in-Loop**!

The "Finger-Stick vs. Venous" Rule for Deploying Enterprise AI

Just as Finger-Stick Edison worked for wide-tolerance **Blood Glucose** ($C_p = 1.43$) but failed on narrow-tolerance **Blood Calcium** ($C_p = 0.61$), current LLMs are **capable** ($C_{pk} \geq 1.33$) **for wide-tolerance tasks** (drafting emails, summarizing notes) but **incapable** ($C_p < 1.0$) **for zero-tolerance autonomous medical dosing** without a deterministic verification wrapper!

Class Discussion: Auditing an AI Triage Bot Using Week 8 SPC

Scenario: Walgreens Deploys an "AI Pharmacy Triage Agent"

- Fast-forward to 2026: Walgreens tests a generative AI voice agent that answers patient medication questions and scores clinical accuracy from 0 to 100 (regulatory minimum LSL $= 85.0$, USL $= 100.0$, target $\mu = 92.5$, width $= 15.0$ points).
- Every hour, an automated evaluator runs the same $n = 5$ "golden reference patient calls" through the AI and logs both the mean score $\bar{X}_t$ and the range $R_t = \max - \min$.

3 Executive Audit Questions

1. **Prompt Tampering:** Every time a single hourly call scores 90.5, an engineer tweaks the AI system prompt by hand. Why will this make the AI *less* reliable?
2. **Hallucination Spike (R-Chart):** On Day 4, the 5 identical test calls score `[72, 93, 94, 99, 92]` ($\bar{X} = 90.0$, $R = 27.0$ vs. baseline $\bar{R} = 4.0$). Why did $\bar{X}$ miss the danger while R caught it?
3. **AI Capability (C_p):** If the LLM's natural score spread is $\sigma = 3.5$ points ($6\sigma = 21.0 > 15.0$ window), can prompt centering alone fix it, or must you lower Temperature / add RAG to shrink σ?

FORMULA REMINDER: Prompt Tampering: $\sigma_{\text{tamp}} = \sqrt{2}\sigma$ | Hallucination Spread: $\text{UCL}_R = D_4\bar{R} = 2.11(4.0) = 8.44$ | AI Capability: $C_p = \frac{15.0}{6(3.5)} = 0.71 < 1.00$

Course Wrap-Up & Complete Week 8 Master Formula Sheet

What We Mastered Across Our 3-Hour Session

- **Blocks 1–4 (Theranos vs. Siemens at Walgreens):**
  - Why knob-twisting on **Glucose** inflates σ by $\sqrt{2}$ ($\pm 3\sigma$ sweet spot).
  - How $\bar{X}$-charts & $n^* = 4$ catch **Potassium** drift (saving \$40k).
  - Why R-charts must be checked *before* $\bar{X}$ on **Vitamin B12**.
  - Why **Blood Calcium** ($C_p = 0.61$) failed while Glucose ($C_p = 1.43$) passed.
- **Block 5 (1999 Belgian Coca-Cola Crisis):** Why a single sensor alarm is 83.8% likely a false alarm when the base rate is 2%, and how $n = 3$ sequential Bayes sampling saves millions.
- **Block 6 (AI in Quality):** Avoiding algorithmic tampering ($\sqrt{2}\sigma$) and auditing LLMs with $\bar{X}$, R, and C_{pk}.

Complete Week 8 Formula Sheet (5 Core Equations!)

1. **Tampering Penalty (Block 1 & AI Block 6):**
   $\sigma_{\text{tampered}} = \sqrt{2}\,\sigma \approx \mathbf{1.414}\sigma \quad (\text{Var} = 2\sigma^2)$
2. **Range Chart First (Block 3 — Vitamin B12):**
   $\text{LCL}_{\mathbf{R}} = \mathbf{D_3}\bar{\mathbf{R}}, \qquad \text{UCL}_{\mathbf{R}} = \mathbf{D_4}\bar{\mathbf{R}}$
3. **Mean Chart Second (Block 2 — Potassium):**
   $\text{LCL}_{\bar{\mathbf{X}}} = \bar{\tilde{\mathbf{X}}} - \mathbf{A_2}\bar{\mathbf{R}}, \qquad \text{UCL}_{\bar{\mathbf{X}}} = \bar{\tilde{\mathbf{X}}} + \mathbf{A_2}\bar{\mathbf{R}}$
4. **Capability Ratios (Block 4 — Blood Calcium):**
   $\mathbf{C_p} = \frac{\text{USL} - \text{LSL}}{\mathbf{6}\sigma}, \quad \mathbf{C_{pk}} = \min\left\{\frac{\text{USL} - \mu}{\mathbf{3}\sigma}, \frac{\mu - \text{LSL}}{\mathbf{3}\sigma}\right\}$
5. **Bayes' Rule Recall Update (Block 5 — Coca-Cola):**
   $\mathbf{P(T \mid D)} = \frac{\mathbf{P(T)P(D \mid T)}}{\mathbf{P(T)P(D \mid T) + P(C)P(D \mid C)}}, \quad \text{Recall if } P(T \mid D) > \frac{C_{\text{recall}}}{C_{\text{harm}}}$

Final Check Before Submitting Canvas Quiz Q1–Q24: Make sure all 24 answers from Rounds 1–4 of `theranos-sim.html` are entered and submitted in Canvas!